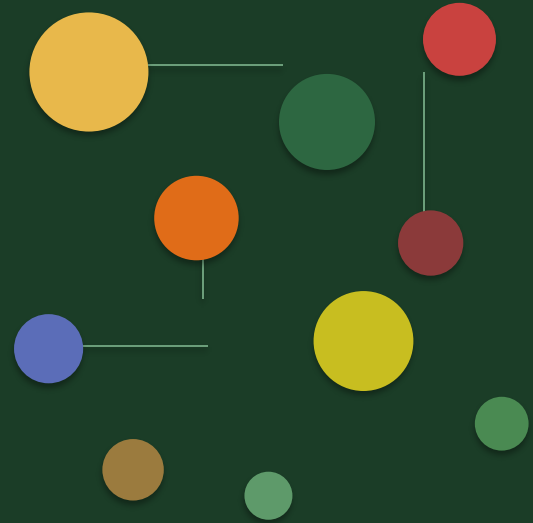


Optimising Product Placement Using Grocery Purchase Data

Turning 3.2 million Instacart orders into assortment,
cross-sell and retention decisions for grocery delivery.

Lara Pestrin · Jonathan Kull · Veprim Mehmeti



Grocery delivery lives or dies by its assortment

Thousands of products.

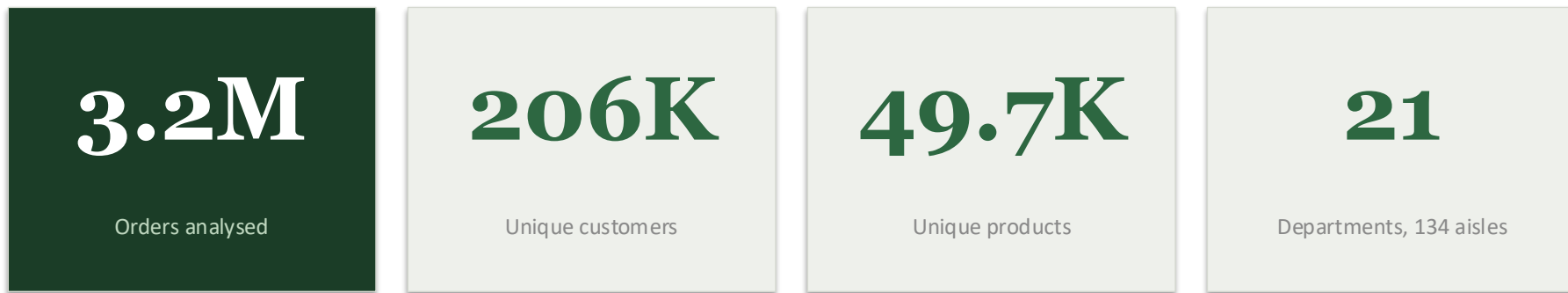
Weekly stocking decisions.

Today — mostly intuition.

THE QUESTIONS WE SET OUT TO ANSWER

- 1 Which products are purchased most frequently?
- 2 Which products are commonly bought together?
- 3 Which products earn the highest customer loyalty?
- 4 When — by day and hour — are orders placed?
- 5 What does this mean for assortment and retention?

Built on 3.2 million real grocery orders



The public Instacart 2017 dataset — six linked tables, fully anonymised.

Two people make these calls every week



Sandra

ASSORTMENT PLANNER

34 · Business Administration · 5 years in grocery retail · no coding

NEEDS

A fast visual overview of what sells — and what sells together.

PAIN

Slow, manual reports; co-purchase patterns are invisible.



Marco

BUYER

28 · Retail Management · 2 years in e-commerce grocery · no coding

NEEDS

Evidence for listing, delisting and supplier negotiations.

PAIN

Decisions lean on gut feel; loyal niche products stay hidden.

An interactive Streamlit dashboard

Built for exploration, not static reporting.

01

KPI Overview

- Orders, customers, products
- Reorder rate & peak hour
- Top department by volume

02

Product Development

- Top sellers by purchase count
- Department sales breakdown
- Aisles bought together
(pairs, heatmap, network)

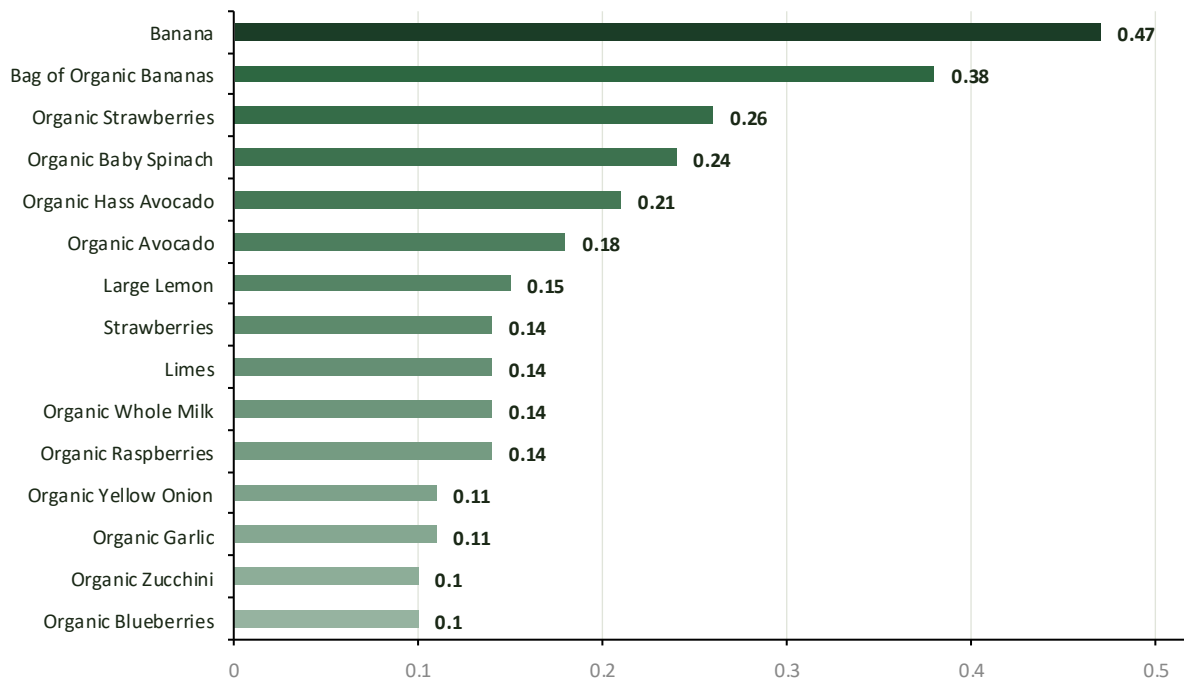
03

Customer Retention

- Reorder rate by product
- Hidden gems — loyal niches
- Shopping-time heatmap

Run it locally: `uv run streamlit run deployment/app.py`

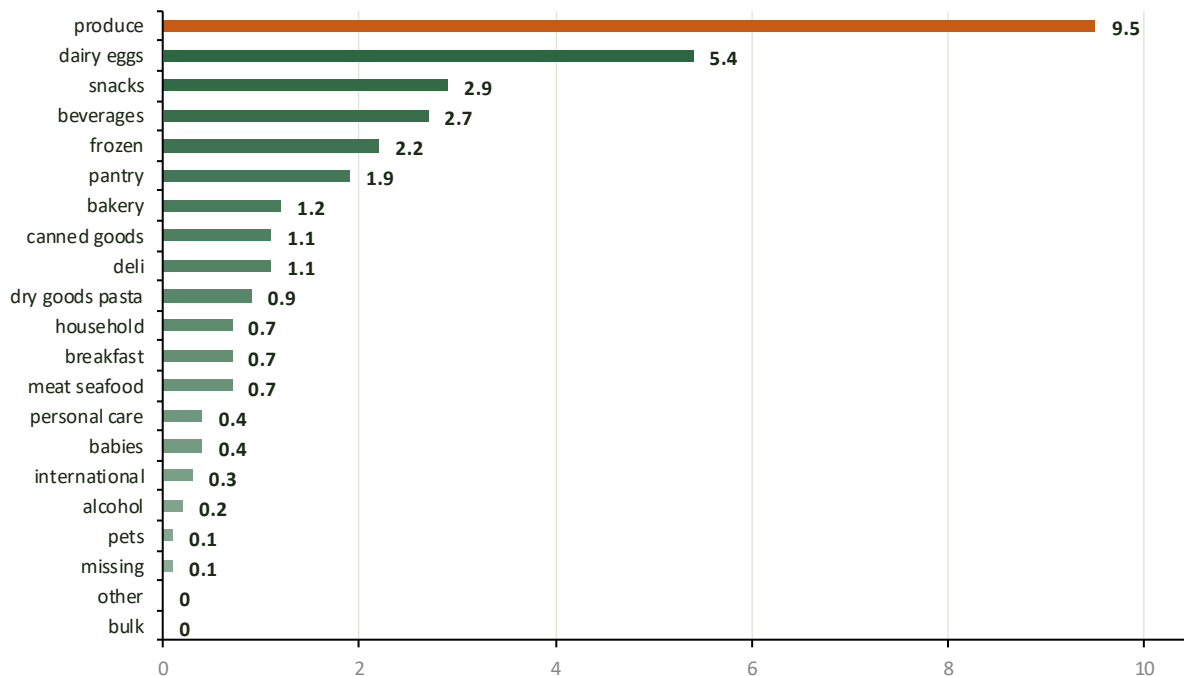
Fresh produce owns the top of the basket



WHAT WE FOUND

Every one of the top 15 products is fresh fruit or vegetables.

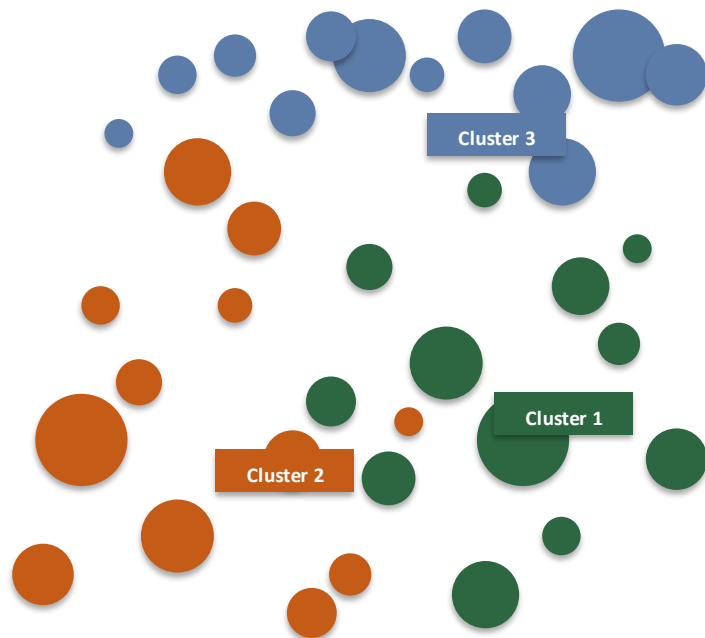
Produce drives almost a third of all volume



WHAT WE FOUND

Produce and Dairy & Eggs dwarf every other department.

Co-purchase analysis reveals natural baskets

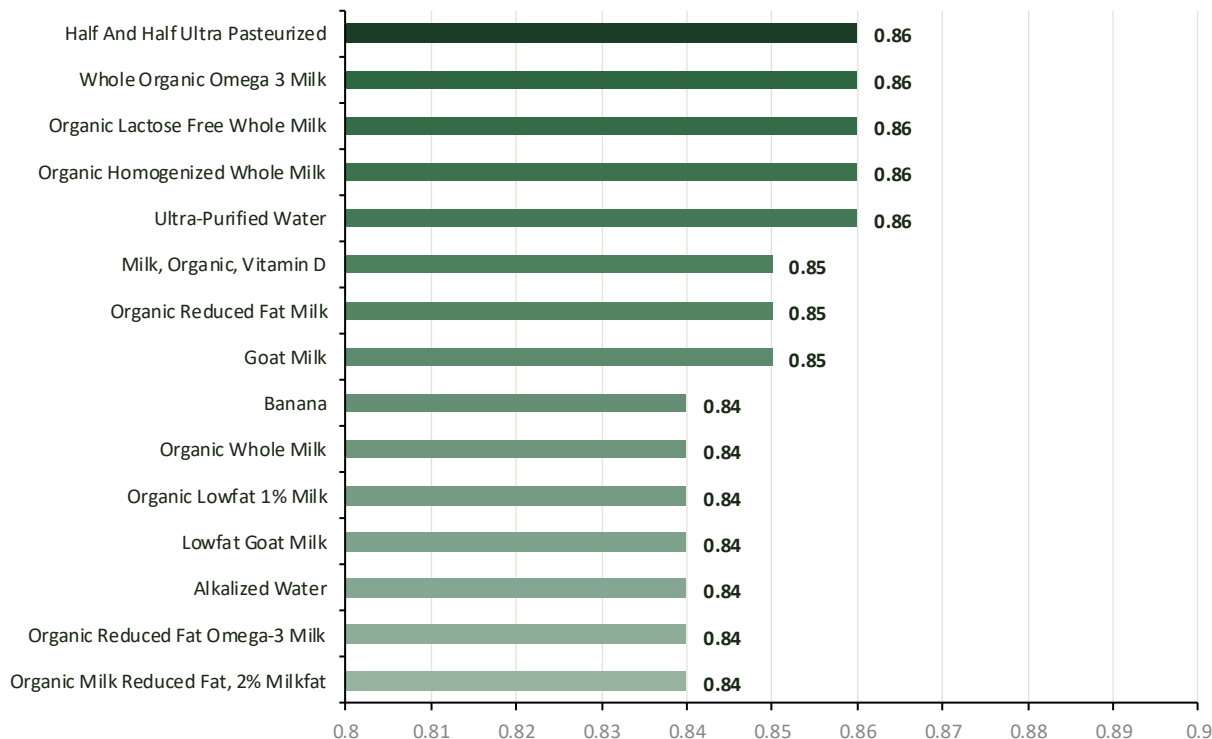


WHAT WE FOUND

Co-purchase patterns fall into three natural baskets.

- Fresh vegetables & aromatics**
spinach, avocado, onion, garlic, lemon
- Everyday fruit & milk**
banana, berries, apples, whole milk
- The organic basket**
organic bananas, berries, almond milk

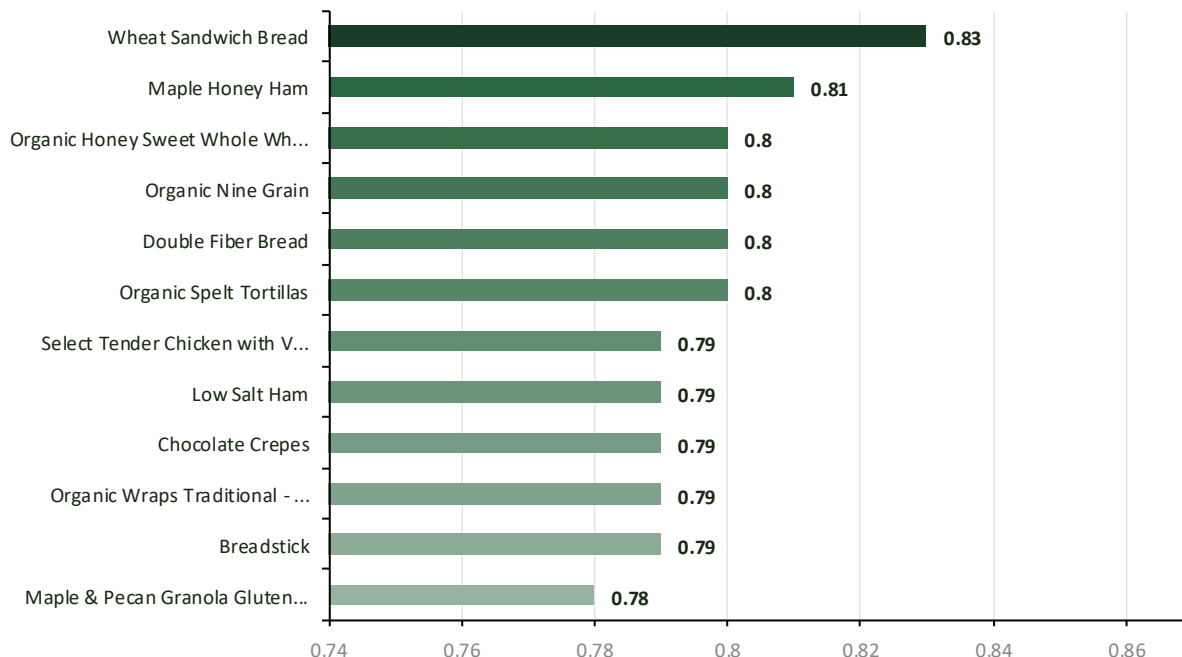
Customer loyalty is built on milk



WHAT WE FOUND

Loyalty runs on milk and water — 84 to 86% reorder.

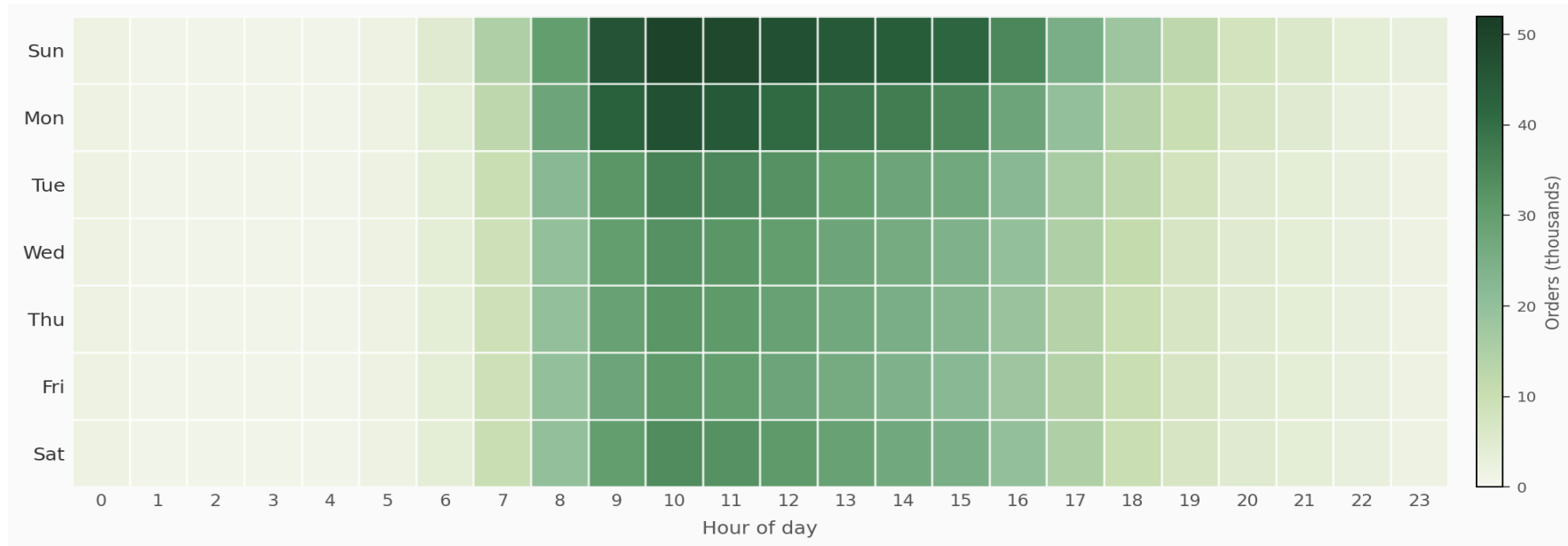
Hidden gems: small categories, loyal fans



WHAT WE FOUND

Niche breads and deli items keep 78 to 83% of their buyers.

Sunday and Monday mornings carry the week



WHAT WE FOUND

Orders cluster on Sunday and Monday, roughly 9:00 to 16:00 — and fall away sharply outside daytime hours.

From data to assortment decisions

01

Build on fresh

Produce and organic are the core.

FOR SANDRA

02

Bundle the clusters

Group and cross-sell what sells together.

FOR MARCO

03

Subscribe the staples

Milk and water — auto-reorder offers.

FOR SANDRA

04

Protect the gems

Keep loyal niche products on the shelf.

FOR MARCO

THANK YOU

Three million orders, one clear shelf strategy.

Anonymised purchase data, turned into decisions Sandra and Marco can act on — without writing a line of code.

Repository github.com/podsv-fs26-ad24/ad24-11-grocery-product-placement-analysis

Run it `uv run streamlit run deployment/app.py`

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